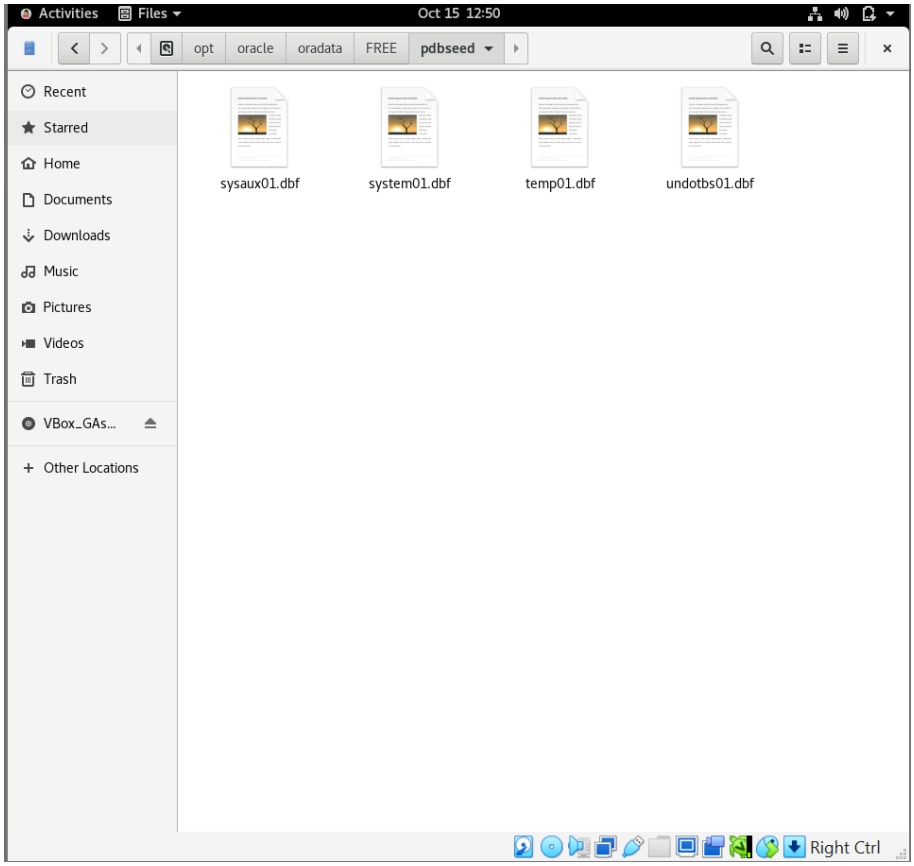
Jocelyn Austin Laurensia

825240044

Creating PDB from seed

Activities - Files - Other Location - Computer

/opt/oracle/oradata/FREE/pdbseed



-- Connect to Oracle database

. oraenv

sqlplus / as sysdba

oracle

```
[oracle@localhost ~]$ . oraenv
ORACLE_SID = [FREE] ?
The Oracle base has been set to /opt/oracle
[oracle@localhost ~]$ sqlplus

SQL*Plus: Release 23.0.0.0.0 - Production on Wed Oct 15 12:52:47 2025
Version 23.9.0.25.07

Copyright (c) 1982, 2025, Oracle.  All rights reserved.

Enter user-name: sys as sysdba
Enter password:

Connected to:
Oracle Database 23ai Free Release 23.0.0.0.0 - Develop, Learn, and Run for Free
Version 23.9.0.25.07

SQL>
```

SQL>show con_name;

```
SQL> show con_name;

CON_NAME
------------------------------
FREEPDB1
```

Menampilkan nama container yang sedang aktif

SQL>col name format a20;

```
SQL> col name format a20;
SQL>
```

Mengatur lebar kolom name menjadi 20 karakter

SQL>select con_id,name,open_mode

from v$containers;

```
SQL> select con_id,name,open_mode
from v$containers;
  2
    CON_ID NAME                 OPEN_MODE
---------- -------------------- ----------
         3 FREEPDB1             READ WRITE

SQL>
```

Menampilkan daftar container beserta status opennya

SQL>alter session set container=PDB$SEED;

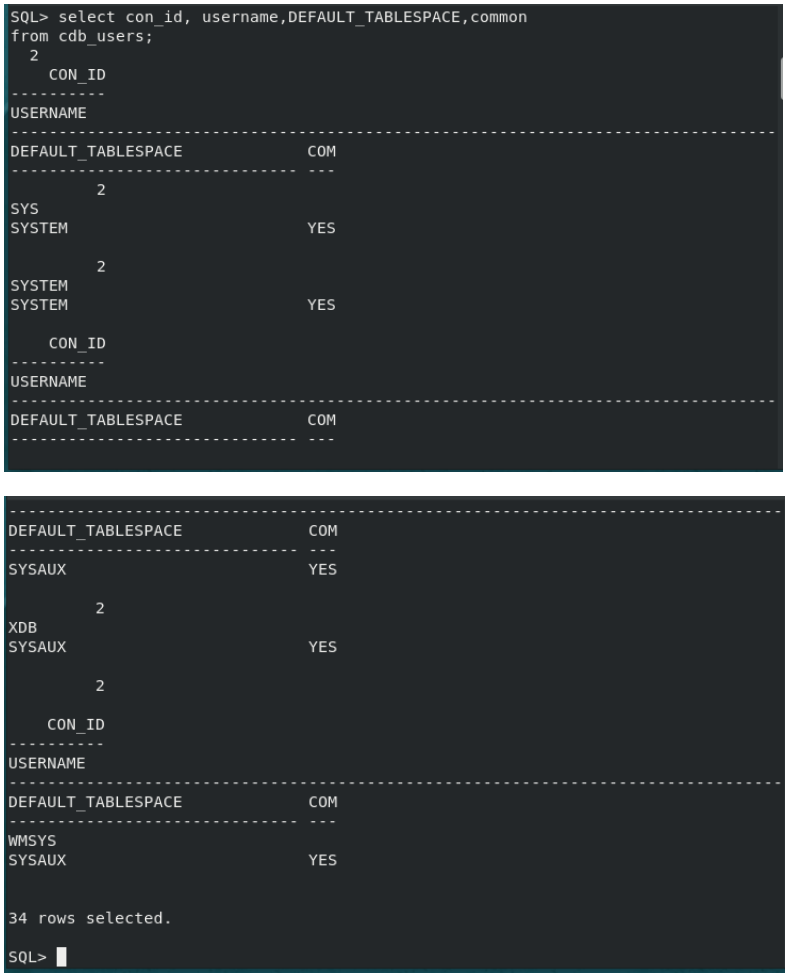
```
SQL> alter session set container=PDB$SEED;

Session altered.
```

Berpindah ke container PDB$SEED

SQL>select con_id, username,DEFAULT_TABLESPACE,common

from cdb_users;



Menampilkan daftar user dari seluruh container beserta default tablespace dan status commonnya dalam CDB_USERS

SQL>select username,DEFAULT_TABLESPACE,common

from dba_users;

```
SQL> select  username,DEFAULT_TABLESPACE,common
from dba_users;
  2
USERNAME
--------------------------------------------------------------------------------
DEFAULT_TABLESPACE             COM
------------------------------ ---
SYS
SYSTEM                         YES

SYSTEM
SYSTEM                         YES

SYSRAC
SYSTEM                         YES


USERNAME
--------------------------------------------------------------------------------
DEFAULT_TABLESPACE             COM
------------------------------ ---
XS$NULL
SYSTEM                         YES
```

```
--------------------------------------------------------------------------------
DEFAULT_TABLESPACE             COM
------------------------------ ---
OLAPSYS
SYSAUX                         YES

MDSYS
SYSAUX                         YES

XDB
SYSAUX                         YES


USERNAME
--------------------------------------------------------------------------------
DEFAULT_TABLESPACE             COM
------------------------------ ---
WMSYS
SYSAUX                         YES


34 rows selected.

SQL>
```

Menampilkan daftar user beserta default tablespace dan status common-nya dalam DBA_USERS

SQL>select con_id,file#, name from V$DATAFILE;

```
SQL> select con_id,file#, name from V$DATAFILE;

    CON_ID      FILE# NAME
---------- ---------- --------------------
         2          2 /opt/oracle/oradata/
                      FREE/pdbseed/system0
                      1.dbf

         2          4 /opt/oracle/oradata/
                      FREE/pdbseed/sysaux0
                      1.dbf

         2          9 /opt/oracle/oradata/
                      FREE/pdbseed/undotbs
                      01.dbf

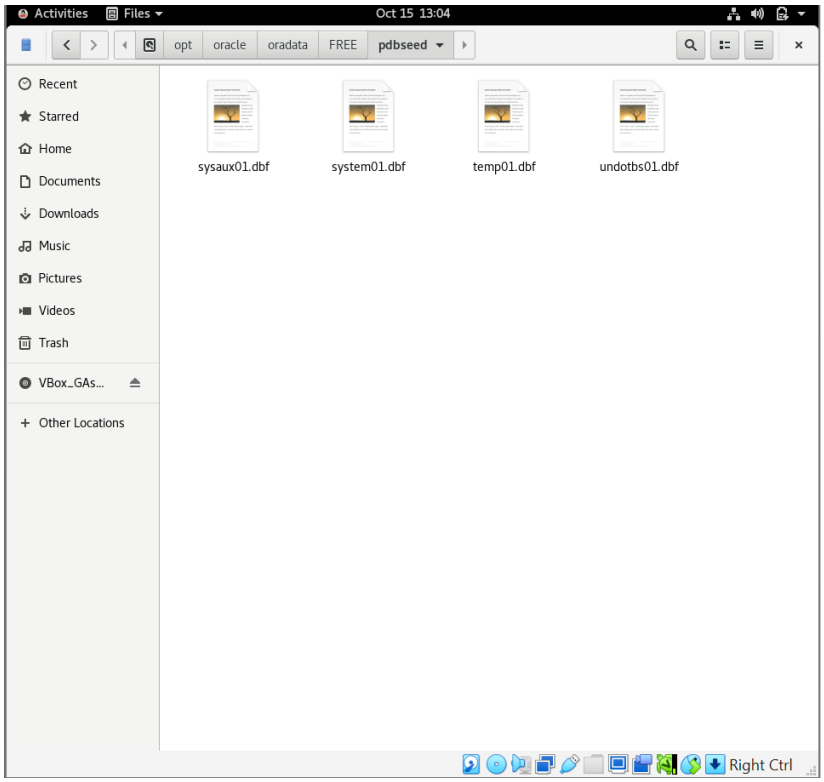
    CON_ID      FILE# NAME
---------- ---------- --------------------


SQL>
```

Menampilkan daftar datafile beserta nomor file dan ID container dalam V$DATAFILE

Activities - Files - Other Location - Computer

opt/oracle/oradata/FREE/pdbseed



SQL>select * from V$TABLESPACE;

```
SQL> select * from V$TABLESPACE;

       TS# NAME                           INC BIG FLA ENC     CON_ID
---------- ------------------------------ --- --- --- --- ----------
         1 SYSAUX                         YES YES YES                2
         0 SYSTEM                         YES YES YES                2
         2 UNDOTBS1                       YES YES YES                2
         3 TEMP                           NO  NO  YES                2

SQL>
```

Menampilkan seluruh informasi tentang tablespace yang ada dalam V$TABLESPACE

SQL>alter session set container=cdb$root;

```
SQL> alter session set container=cdb$root;

Session altered.
```

Berpindah ke container CDB$ROOT

-- mkdir /opt/oracle/oradata/FREE/pdb1

atau langsung dari SQL:

SQL> !mkdir /opt/oracle/oradata/FREE/pdb1

```
SQL> !mkdir /opt/oracle/oradata/FREE/pdb1

SQL>
```

Membuat direktori /opt/oracle/oradata/FREE/pdb1 untuk menyimpan file data PDB baru

SQL>CREATE PLUGGABLE DATABASE pdb1

ADMIN USER pdb1admin IDENTIFIED BY welcome

ROLES = (dba)

DEFAULT TABLESPACE users

DATAFILE '/opt/oracle/oradata/FREE/pdb1users01.dbf' SIZE 250M AUTOEXTEND ON

FILE_NAME_CONVERT = ('/opt/oracle/oradata/FREE/pdbseed/',

'/opt/oracle/oradata/FREE/pdb1/');

```
SQL> CREATE PLUGGABLE DATABASE pdb1
  ADMIN USER pdb1admin IDENTIFIED BY welcome
  ROLES = (dba)
  DEFAULT TABLESPACE users
    DATAFILE '/opt/oracle/oradata/FREE/pdb1users01.dbf' SIZE 250M AUTOEXTEND ON
  FILE_NAME_CONVERT = ('/opt/oracle/oradata/FREE/pdbseed/',
                       '/opt/oracle/oradata/FREE/pdb1/');
  2    3    4    5    6    7  CREATE PLUGGABLE DATABASE pdb1
*
ERROR at line 1:
ORA-01017: invalid credential or not authorized; logon denied
Help: https://docs.oracle.com/error-help/db/ora-01017/


SQL>
```

Error karena path pada FILE_NAME_CONVERT tidak sesuai dengan lokasi sebenarnya dari pdbseed

SQL>select con_id,name,open_mode

from v$containers;

```
SQL> select con_id,name,open_mode
from v$containers;
  2
    CON_ID NAME                 OPEN_MODE
---------- -------------------- ----------
         1 CDB$ROOT             READ WRITE
         2 PDB$SEED             READ ONLY
         3 FREEPDB1             READ WRITE
```

Menampilkan container id, nama dan status opennya dari V$CONTAINERS

SQL>alter session set container=pdb1;

```
SQL> alter session set container=pdb1;
ERROR:
ORA-65011: Pluggable database PDB1 does not exist.
Help: https://docs.oracle.com/error-help/db/ora-65011/


SQL>
```

Berpindah ke container PDB1

SQL>alter pluggable database open;

```
SQL> alter pluggable database open;
alter pluggable database open
*
ERROR at line 1:
ORA-65000: missing or invalid pluggable database name
Help: https://docs.oracle.com/error-help/db/ora-65000/


SQL>
```

Error karena tidak menyebutkan secara spesifik PDB mana yang ingin dibuka

SQL>select con_id, username,DEFAULT_TABLESPACE,common

from cdb_users;

```
SQL> select con_id, username,DEFAULT_TABLESPACE,common
from cdb_users;
  2
    CON_ID
----------
USERNAME
--------------------------------------------------------------------------------
DEFAULT_TABLESPACE             COM
------------------------------ ---
         1
SYS
SYSTEM                         YES

         1
SYSTEM
SYSTEM                         YES

    CON_ID
----------
USERNAME
--------------------------------------------------------------------------------
DEFAULT_TABLESPACE             COM
```

```
DEFAULT_TABLESPACE             COM
------------------------------ ---
SYSKM
USERS                          YES

         3
SYS$UMF
USERS                          YES


    CON_ID
----------
USERNAME
--------------------------------------------------------------------------------
DEFAULT_TABLESPACE             COM
------------------------------ ---
         3
SYSDG
USERS                          YES


71 rows selected.

SQL>
```

Menampilkan container id, username, default tablespace dan status commonnya dalam CDB_USERS

SQL> col username format a20;

```
SQL> col username format a20;
SQL>
```

Mengatur lebar kolom username menjadi 20 karakter

SQL>select username,DEFAULT_TABLESPACE,common

from dba_users;

```
SQL> select  username,DEFAULT_TABLESPACE,common
from dba_users;
  2
USERNAME             DEFAULT_TABLESPACE             COM
-------------------- ------------------------------ ---
SYS                  SYSTEM                         YES
SYSTEM               SYSTEM                         YES
SYSRAC               USERS                          YES
XS$NULL              SYSTEM                         YES
OJVMSYS              SYSTEM                         YES
LBACSYS              SYSTEM                         YES
OUTLN                SYSTEM                         YES
DBSNMP               SYSAUX                         YES
APPQOSSYS            SYSAUX                         YES
VECSYS               SYSAUX                         YES
DBSFWUSER            SYSAUX                         YES

USERNAME             DEFAULT_TABLESPACE             COM
-------------------- ------------------------------ ---
GGSYS                SYSAUX                         YES
ANONYMOUS            SYSAUX                         YES
CTXSYS               SYSAUX                         YES
```

```
USERNAME             DEFAULT_TABLESPACE             COM
-------------------- ------------------------------ ---
XDB                  SYSAUX                         YES
WMSYS                SYSAUX                         YES
GSMCATUSER           USERS                          YES
MDDATA               USERS                          YES
REMOTE_SCHEDULER_AGE USERS                          YES
NT

SYSBACKUP            USERS                          YES
GSMUSER              USERS                          YES
GSMROOTUSER          USERS                          YES
DIP                  USERS                          YES

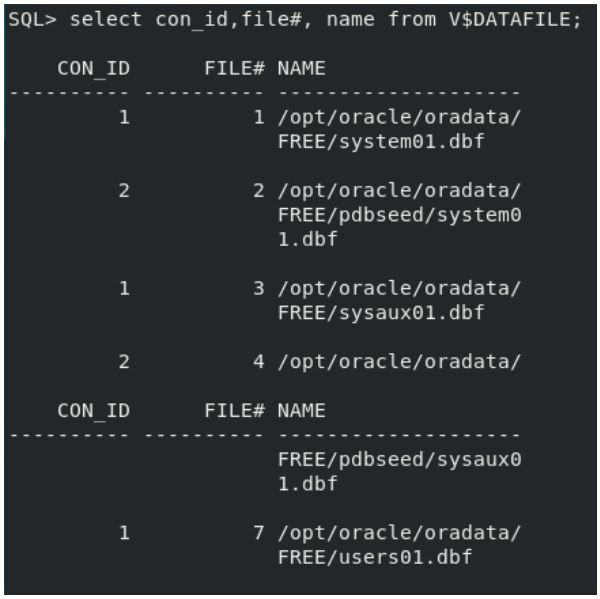
USERNAME             DEFAULT_TABLESPACE             COM
-------------------- ------------------------------ ---
DGPDB_INT            USERS                          YES
SYSKM                USERS                          YES
SYS$UMF              USERS                          YES
SYSDG                USERS                          YES

35 rows selected.

SQL>
```

Menampilkan user beserta default tablespace dan status commonnya dalam DBA_USERS

SQL>select con_id,file#, name from V$DATAFILE;



```
SQL> select con_id,file#, name from V$DATAFILE;

    CON_ID      FILE# NAME
---------- ---------- --------------------
         1          1 /opt/oracle/oradata/
                      FREE/system01.dbf

         2          2 /opt/oracle/oradata/
                      FREE/pdbseed/system0
                      1.dbf

         1          3 /opt/oracle/oradata/
                      FREE/sysaux01.dbf

         2          4 /opt/oracle/oradata/

    CON_ID      FILE# NAME
---------- ---------- --------------------
                      FREE/pdbseed/sysaux0
                      1.dbf

         1          7 /opt/oracle/oradata/
                      FREE/users01.dbf
```



```
         3         12 /opt/oracle/oradata/
                      FREE/FREEPDB1/system
                      01.dbf

         3         13 /opt/oracle/oradata/
                      FREE/FREEPDB1/sysaux
                      01.dbf

         3         14 /opt/oracle/oradata/

    CON_ID      FILE# NAME
---------- ---------- --------------------
                      FREE/FREEPDB1/undotb
                      s01.dbf

         3         15 /opt/oracle/oradata/
                      FREE/FREEPDB1/users0
                      1.dbf


11 rows selected.

SQL>
```

Menampilkan daftar datafile beserta nomor file dan ID container dalam V$DATAFILE

SQL>select * from V$TABLESPACE;

```
SQL> select * from V$TABLESPACE;

       TS# NAME                 INC BIG FLA ENC     CON_ID
---------- -------------------- --- --- --- --- ----------
         1 SYSAUX               YES YES YES              1
         0 SYSTEM               YES YES YES              1
         2 UNDOTBS1             YES YES YES              1
         4 USERS                YES YES YES              1
         1 SYSAUX               YES YES YES              2
         0 SYSTEM               YES YES YES              2
         2 UNDOTBS1             YES YES YES              2
         3 TEMP                 NO  NO  YES              1
         3 TEMP                 NO  NO  YES              2
         0 SYSTEM               YES YES YES              3
         1 SYSAUX               YES YES YES              3

       TS# NAME                 INC BIG FLA ENC     CON_ID
---------- -------------------- --- --- --- --- ----------
         2 UNDOTBS1             YES YES YES              3
         3 TEMP                 NO  NO  YES              3
         6 USERS                YES YES YES              3
```

```
       TS# NAME                 INC BIG FLA ENC     CON_ID
---------- -------------------- --- --- --- --- ----------
         1 SYSAUX               YES YES YES              1
         0 SYSTEM               YES YES YES              1
         2 UNDOTBS1             YES YES YES              1
         4 USERS                YES YES YES              1
         1 SYSAUX               YES YES YES              2
         0 SYSTEM               YES YES YES              2
         2 UNDOTBS1             YES YES YES              2
         3 TEMP                 NO  NO  YES              1
         3 TEMP                 NO  NO  YES              2
         0 SYSTEM               YES YES YES              3
         1 SYSAUX               YES YES YES              3

       TS# NAME                 INC BIG FLA ENC     CON_ID
---------- -------------------- --- --- --- --- ----------
         2 UNDOTBS1             YES YES YES              3
         3 TEMP                 NO  NO  YES              3
         6 USERS                YES YES YES              3

14 rows selected.

SQL>
```

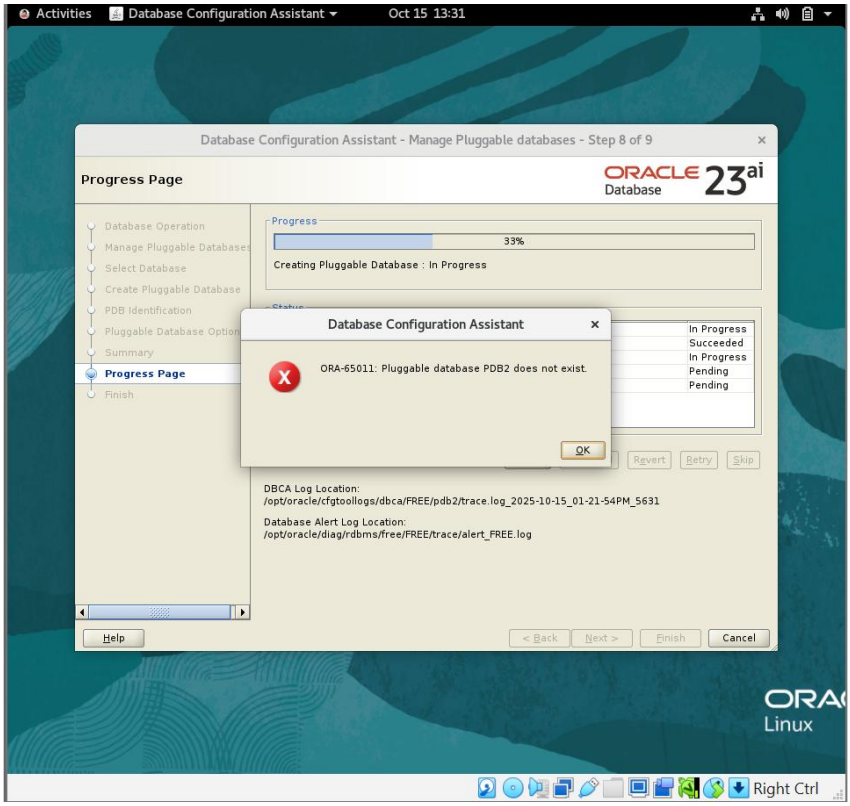
Menampilkan seluruh informasi mengenai tablespace melalui V$TABLESPACE

Creating PDB from seed (DBCA)

-- Open new Terminal

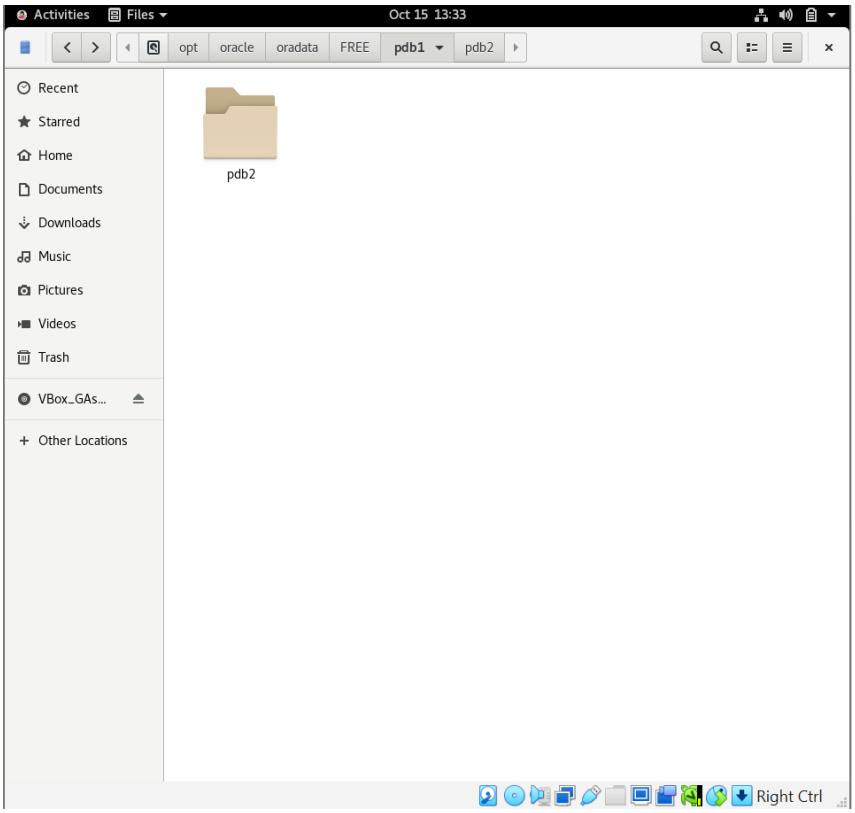
. oraenv

$ dbca



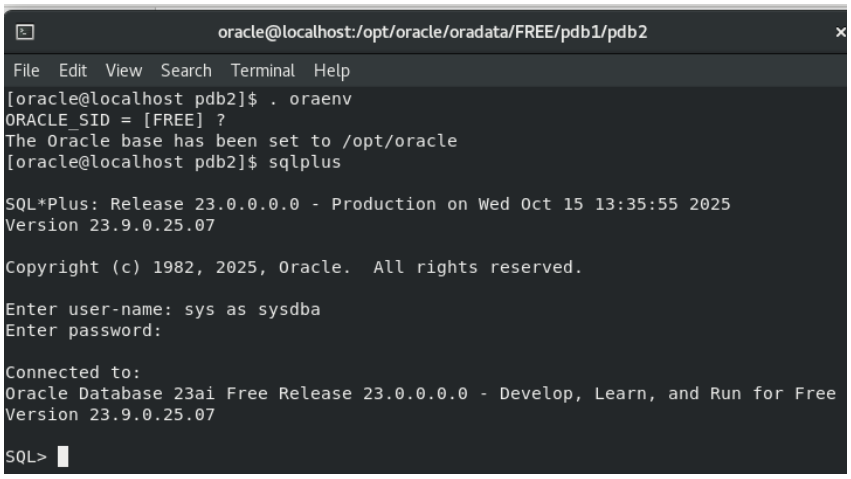
Activities - Files - Other Locations - Computer

/opt/oracle/oradata/FREE/pdb2/

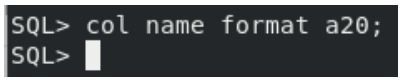


Right click - Open in Terminal

sqlplus / as sysdba

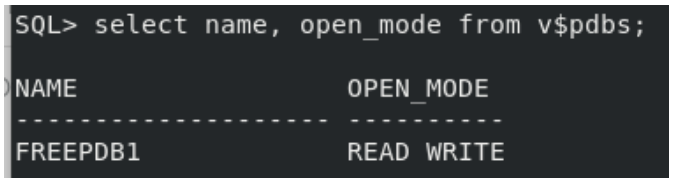


SQL> col name format a20;



Mengatur lebar kolom name menjadi 20 karakter

SQL> select name, open_mode from v$pdbs;



Menampilkan nama setiap PDB beserta status open-nya dari V$PDBS

Dropping pluggable using SQL

Activities - Files - Other Locations - Computer

/opt/oracle/oradata/FREE

Right click - Open in Terminal

sqlplus / as sysdba

```
[oracle@localhost FREE]$ sqlplus

SQL*Plus: Release 23.0.0.0.0 - Production on Wed Oct 15 13:39:29 2025
Version 23.9.0.25.07

Copyright (c) 1982, 2025, Oracle.  All rights reserved.

Enter user-name: sys as sysdba
Enter password:

Connected to:
Oracle Database 23ai Free Release 23.0.0.0.0 - Develop, Learn, and Run for Free
Version 23.9.0.25.07

SQL>
```

SQL> col name format a20;

```
SQL> col name format a20;
SQL>
```

Mengatur lebar kolom name menjadi 20 karakter

SQL> select name, open_mode from v$pdbs;

```
SQL> select name, open_mode from v$pdbs;

NAME                 OPEN_MODE
-------------------- ----------
FREEPDB1             READ WRITE

SQL>
```

Menampilkan daftar nama PDB beserta status opennya dalam V$PDBS

SQL> drop pluggable database pdb2 including datafiles;

```
SQL> drop pluggable database pdb2 including datafiles;
drop pluggable database pdb2 including datafiles
*
ERROR at line 1:
ORA-65040: Operation is not allowed from within a pluggable database.
Help: https://docs.oracle.com/error-help/db/ora-65040/


SQL>
```

error terjadi karena PDB2 tidak ada dalam daftar PDB

```
SQL> alter session set container = cdb$root;

Session altered.

SQL>
```

SQL> alter pluggable database pdb2 close;

```
SQL> alter pluggable database pdb2 close;
alter pluggable database pdb2 close
*
ERROR at line 1:
ORA-65011: Pluggable database PDB2 does not exist.
Help: https://docs.oracle.com/error-help/db/ora-65011/


SQL>
```

Error karena PDB2 belum ada di dalam container database, sehingga tidak bisa ditutup

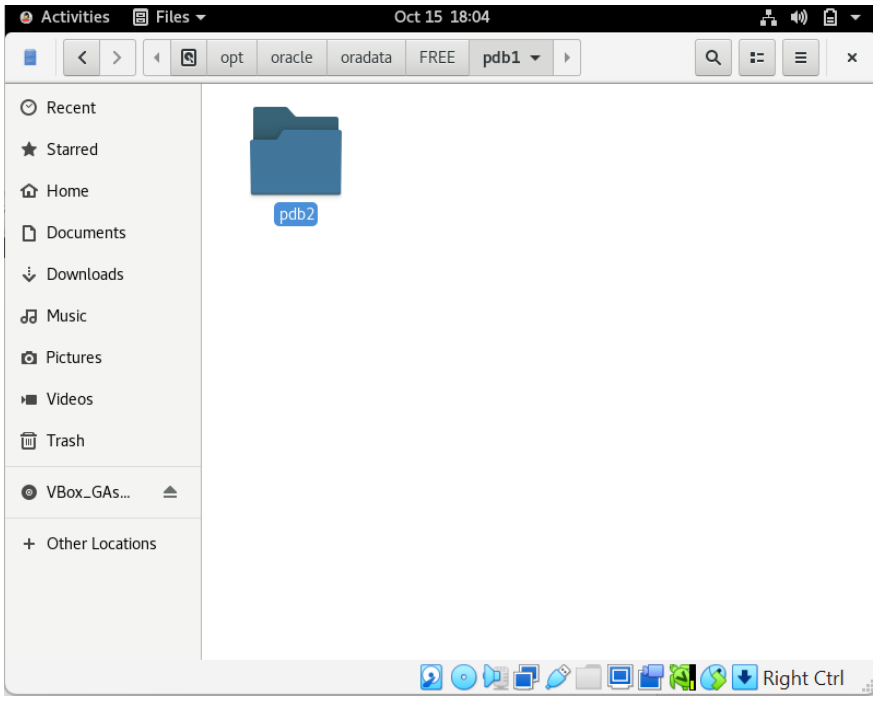
SQL> drop pluggable database pdb2 including datafiles;

```
SQL> drop pluggable database pdb2 including datafiles;
drop pluggable database pdb2 including datafiles
*
ERROR at line 1:
ORA-01017: invalid credential or not authorized; logon denied
Help: https://docs.oracle.com/error-help/db/ora-01017/
```

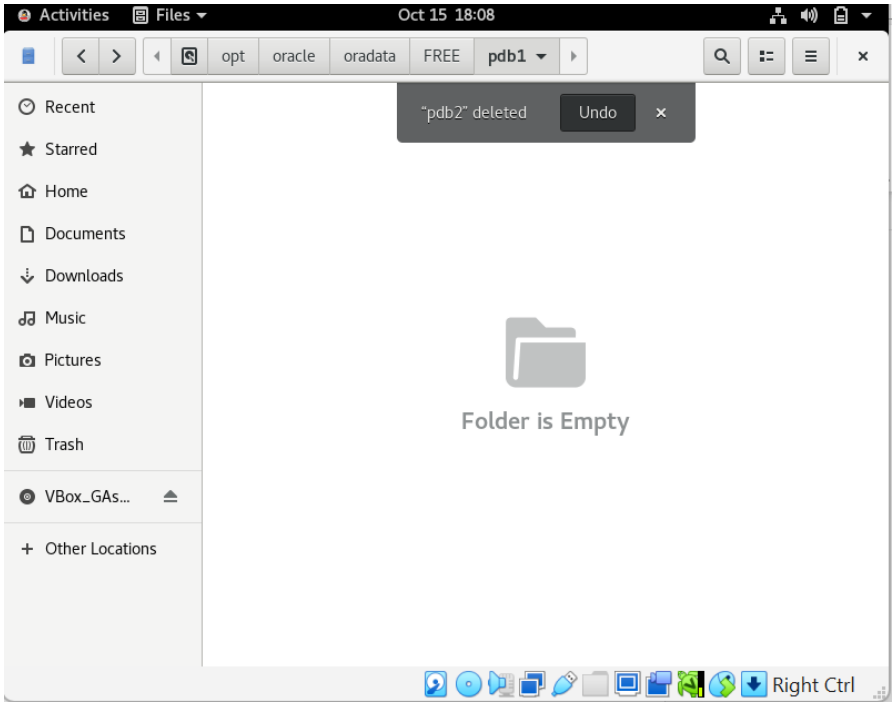
Error karena PDB2 belum pernah dibuat, sehingga tidak ada database bernama PDB2 untuk dihapus

Activities - Files - Other Locations - Computer

/opt/oracle/oradata/FREE/pdb2



right click directory pdb2 - Delete Permanently



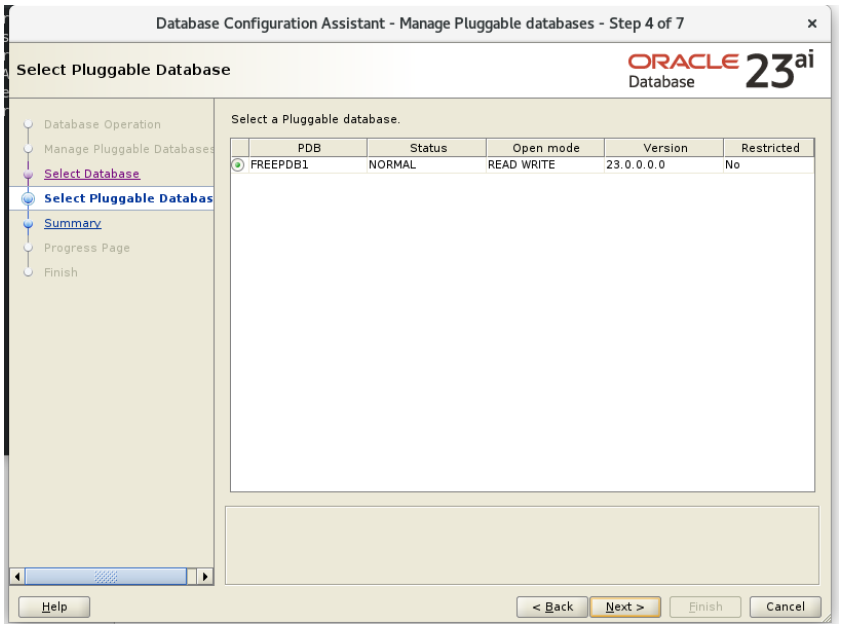
Dropping pluggable using DBCA

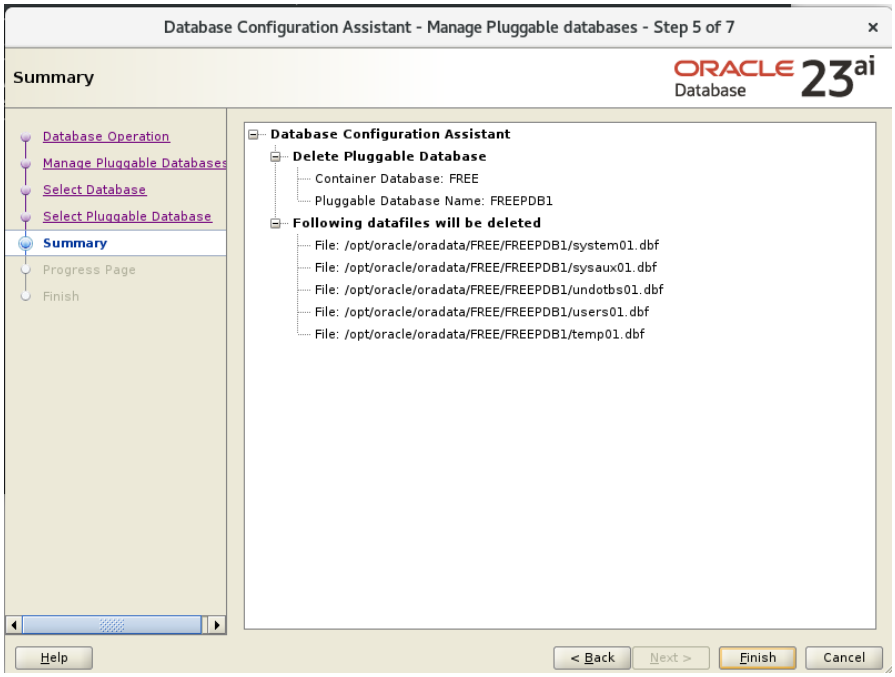
Activities - Files - Other Locations - Computer

/opt/oracle/oradata/FREE/

Right click - Open in Terminal

$ dbca





Cloning PDBs

SQL> show con_name;

```
SQL> show con_name;

CON_NAME
------------------------------
FREEPDB1
SQL>
```

Menampilkan nama container database yang sedang aktif

SQL> col name format a20;

```
SQL> col name format a20;
SQL>
```

Mengatur lebar kolom name menjadi 20 karakter

SQL> select con_id,name,open_mode

from v$pdbs;

```
SQL> select con_id,name,open_mode
from v$pdbs;
  2
    CON_ID NAME                 OPEN_MODE
---------- -------------------- ----------
         3 FREEPDB1             READ WRITE

SQL>
```

Menampilkan daftar PDB, ID container dan status opennya dari V$PDBS

SQL> alter pluggable database all open;

```
SQL> alter pluggable database all open;
alter pluggable database all open
*
ERROR at line 1:
ORA-65040: Operation is not allowed from within a pluggable database.
Help: https://docs.oracle.com/error-help/db/ora-65040/


SQL>
```

```
SQL> alter session set container=cdb$root;

Session altered.

SQL> alter pluggable database all open;

Pluggable database altered.
```

Error terjadi karena perintah dijalankan bukan dari container utama (CDB$ROOT) sehingga perlu melakukan alter session ke cdb$root

SQL> select con_id,name,open_mode

from v$pdbs;

```
SQL> select con_id,name,open_mode
from v$pdbs;
  2
    CON_ID NAME                 OPEN_MODE
---------- -------------------- ----------
         2 PDB$SEED             READ ONLY
         3 FREEPDB1             READ WRITE

SQL>
```

Menampilkan daftar pluggable database (PDB), ID container dan status opennya dari V$PDBS

SQL> alter session set container=FREEPDB1;

```
SQL> alter session set container=FREEPDB1;

Session altered.

SQL>
```

Berpindah ke container FREEPDB1 agar perintah berikutnya dijalankan di dalam PDB

SQL> show con_name;

```
SQL> show con_name;

CON_NAME
------------------------------
FREEPDB1
SQL>
```

Menampilkan nama container database yang sedang aktif

SQL> select username,DEFAULT_TABLESPACE,common

from dba_users;

```
SQL> select username,DEFAULT_TABLESPACE,common
from dba_users;
  2
USERNAME
--------------------------------------------------------------------------------
DEFAULT_TABLESPACE             COM
------------------------------ ---
SYS
SYSTEM                         YES

SYSTEM
SYSTEM                         YES

PDBADMIN
USERS                          NO


USERNAME
--------------------------------------------------------------------------------
DEFAULT_TABLESPACE             COM
------------------------------ ---
SYSRAC
USERS                          YES
```

```
DIP
USERS                          YES

DGPDB_INT
USERS                          YES


USERNAME
--------------------------------------------------------------------------------
DEFAULT_TABLESPACE             COM
------------------------------ ---
SYSKM
USERS                          YES

SYS$UMF
USERS                          YES

SYSDG
USERS                          YES


36 rows selected.

SQL>
```

Menampilkan daftar user, default tablespace dan status commonnya dalam DBA_USERS

SQL> select con_id,file#, name from V$DATAFILE;

```
SQL> select con_id,file#, name from V$DATAFILE;

    CON_ID      FILE# NAME
---------- ---------- --------------------
         3         12 /opt/oracle/oradata/
                      FREE/FREEPDB1/system
                      01.dbf

         3         13 /opt/oracle/oradata/
                      FREE/FREEPDB1/sysaux
                      01.dbf

         3         14 /opt/oracle/oradata/
                      FREE/FREEPDB1/undotb
                      s01.dbf

    CON_ID      FILE# NAME
---------- ---------- --------------------

         3         15 /opt/oracle/oradata/
                      FREE/FREEPDB1/users0
                      1.dbf
```

Menampilkan daftar datafile, nomor file dan ID container dalam V$DATAFILE

SQL> select * from V$TABLESPACE;

```
SQL> select  * from V$TABLESPACE;

       TS# NAME                 INC BIG FLA ENC     CON_ID
---------- -------------------- --- --- --- --- ----------
         0 SYSTEM               YES YES YES              3
         1 SYSAUX               YES YES YES              3
         2 UNDOTBS1             YES YES YES              3
         3 TEMP                 NO  NO  YES              3
         6 USERS                YES YES YES              3

SQL>
```

Menampilkan seluruh informasi tablespace melalui V$TABLESPACE

SQL> !mkdir /opt/oracle/oradata/FREE/pdb5

```
SQL> !mkdir /opt/oracle/oradata/FREE/pdb5

SQL>
```

Membuat direktori /opt/oracle/oradata/FREE/pdb5 untuk menyimpan file data PDB

SQL> alter session set container=cdb$root;

```
SQL> alter session set container=cdb$root;

Session altered.

SQL>
```

Berpindah ke container utama CDB$ROOT

SQL> show con_name;

```
SQL> show con_name;

CON_NAME
------------------------------
CDB$ROOT
SQL>
```

Menampilkan nama container database yang sedang aktif

SQL> CREATE PLUGGABLE DATABASE pdb5 from freepdb1

FILE_NAME_CONVERT = ('/opt/oracle/oradata/FREE/freepdb1/',

'/opt/oracle/oradata/FREE/pdb5/');

```
SQL> CREATE PLUGGABLE DATABASE pdb5 from freepdb1
FILE_NAME_CONVERT = ('/opt/oracle/oradata/FREE/freepdb1/',
                     '/opt/oracle/oradata/FREE/pdb5/');
  2    3  CREATE PLUGGABLE DATABASE pdb5 from freepdb1
*
ERROR at line 1:
ORA-01017: invalid credential or not authorized; logon denied
Help: https://docs.oracle.com/error-help/db/ora-01017/


SQL>
```

Error terjadi karena perintah dijalankan bukan dari CDB$ROOT dan pembuatan folder tidak sama dengan pembuatan PDB, jadi PDB5 belum terdaftar di database meskipun direktori /pdb5 sudah dibuat

SQL> select con_id,name,open_mode

from v$pdbs;

```
SQL> select con_id,name,open_mode
from v$pdbs;
  2
    CON_ID NAME                 OPEN_MODE
---------- -------------------- ----------
         2 PDB$SEED             READ ONLY
         3 FREEPDB1             READ WRITE

SQL>
```

Menampilkan daftar pluggable database (PDB), ID container dan status opennya dari V$PDBS

SQL> alter session set container=PDB5;

```
SQL> alter session set container=PDB5;
ERROR:
ORA-65011: Pluggable database PDB5 does not exist.
Help: https://docs.oracle.com/error-help/db/ora-65011/


SQL>
```

Error karena PDB5 belum dibuat, sehingga container PDB5 tidak ditemukan

SQL> alter pluggable database open;

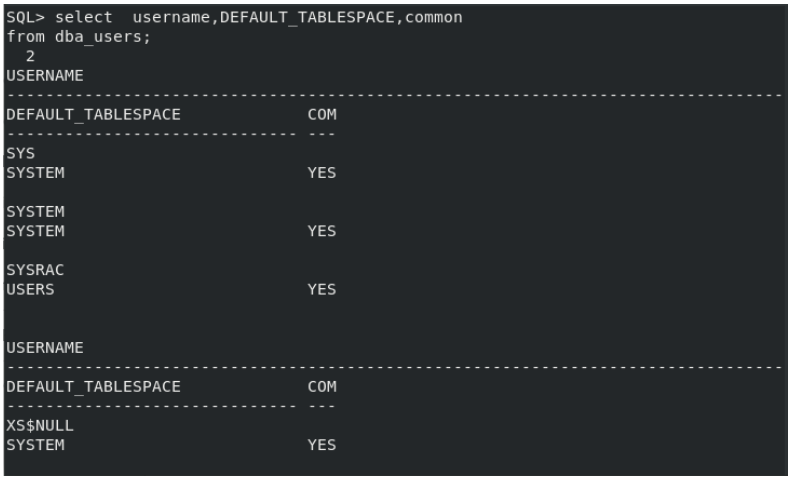
```
SQL> alter pluggable database open;
alter pluggable database open
*
ERROR at line 1:
ORA-65000: missing or invalid pluggable database name
Help: https://docs.oracle.com/error-help/db/ora-65000/


SQL>
```

Error karena perintah dijalankan tanpa menyebutkan nama PDB yang ingin dibuka

SQL> select username,DEFAULT_TABLESPACE,common

from dba_users;

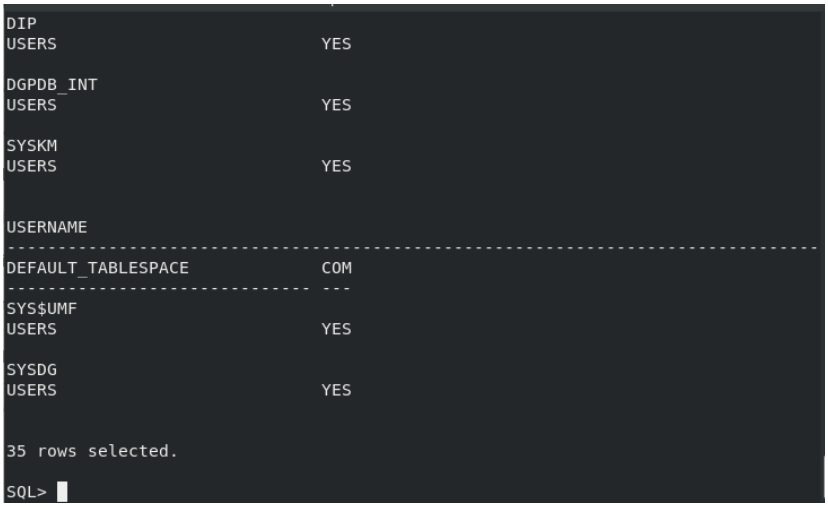


```
SQL> select  username,DEFAULT_TABLESPACE,common
from dba_users;
  2
USERNAME
--------------------------------------------------------------------------------
DEFAULT_TABLESPACE             COM
------------------------------ ---
SYS
SYSTEM                         YES

SYSTEM
SYSTEM                         YES

SYSRAC
USERS                          YES


USERNAME
--------------------------------------------------------------------------------
DEFAULT_TABLESPACE             COM
------------------------------ ---
XS$NULL
SYSTEM                         YES
```



```
DIP
USERS                           YES

DGPDB_INT
USERS                           YES

SYSKM
USERS                           YES


USERNAME
--------------------------------------------------------------------------------
DEFAULT_TABLESPACE             COM
------------------------------ ---
SYS$UMF
USERS                          YES

SYSDG
USERS                          YES


35 rows selected.

SQL>
```

Menampilkan daftar user, default tablespace dan status commonnya dalam DBA_USERS

SQL> select con_id, file#, name from V$DATAFILE;

```
SQL> select con_id, file#, name from V$DATAFILE;

    CON_ID      FILE# NAME
---------- ---------- --------------------
         1          1 /opt/oracle/oradata/
                      FREE/system01.dbf

         2          2 /opt/oracle/oradata/
                      FREE/pdbseed/system0
                      1.dbf

         1          3 /opt/oracle/oradata/
                      FREE/sysaux01.dbf

         2          4 /opt/oracle/oradata/

    CON_ID      FILE# NAME
---------- ---------- --------------------
                      FREE/pdbseed/sysaux0
                      1.dbf

         1          7 /opt/oracle/oradata/
                      FREE/users01.dbf
```

```
         3         12 /opt/oracle/oradata/
                      FREE/FREEPDB1/system
                      01.dbf

         3         13 /opt/oracle/oradata/
                      FREE/FREEPDB1/sysaux
                      01.dbf

         3         14 /opt/oracle/oradata/

    CON_ID      FILE# NAME
---------- ---------- --------------------
                      FREE/FREEPDB1/undotb
                      s01.dbf

         3         15 /opt/oracle/oradata/
                      FREE/FREEPDB1/users0
                      1.dbf


11 rows selected.

SQL>
```

Menampilkan daftar datafile, nomor file dan ID container dalam V$DATAFILE

SQL> select * from V$TABLESPACE;

```
SQL> select * from V$TABLESPACE;

       TS# NAME                 INC BIG FLA ENC     CON_ID
---------- -------------------- --- --- --- --- ----------
         1 SYSAUX               YES YES YES              1
         0 SYSTEM               YES YES YES              1
         2 UNDOTBS1             YES YES YES              1
         4 USERS                YES YES YES              1
         1 SYSAUX               YES YES YES              2
         0 SYSTEM               YES YES YES              2
         2 UNDOTBS1             YES YES YES              2
         3 TEMP                 NO  NO  YES              1
         3 TEMP                 NO  NO  YES              2
         0 SYSTEM               YES YES YES              3
         1 SYSAUX               YES YES YES              3

       TS# NAME                 INC BIG FLA ENC     CON_ID
---------- -------------------- --- --- --- --- ----------
         2 UNDOTBS1             YES YES YES              3
         3 TEMP                 NO  NO  YES              3
         6 USERS                YES YES YES              3

14 rows selected.
```

Menampilkan seluruh informasi tablespace melalui V$TABLESPACE

Unplugging and plugging in PDBs

Activities - Files - Other Locations - Computer

/opt/oracle/oradata/FREE

SQL> show con_name

```
SQL> show con_name

CON_NAME
------------------------------
CDB$ROOT
SQL>
```

Menampilkan nama container yang sedang aktif

SQL> alter pluggable database all open;

```
SQL> alter pluggable database all open;

Pluggable database altered.

SQL>
```

Membuka semua pluggable database (PDB) dari CDB$ROOT

SQL> col name format a20;

```
SQL> col name format a20;
SQL>
```

Mengatur lebar kolom menjadi 20 karakter

SQL> select con_id, name, open_mode

from v$pdbs;

```
SQL> select con_id, name, open_mode
from v$pdbs;
  2
    CON_ID NAME                 OPEN_MODE
---------- -------------------- ----------
         2 PDB$SEED             READ ONLY
         3 FREEPDB1             READ WRITE

SQL>
```

Menampilkan daftar PDB beserta ID container dan status opennya dari V$PDBS

SQL> alter session set container=pdb5;

```
SQL> alter session set container=pdb5;
ERROR:
ORA-65011: Pluggable database PDB5 does not exist.
Help: https://docs.oracle.com/error-help/db/ora-65011/


SQL>
```

Error karena PDB5 belum dibuat, sehingga container PDB5 tidak ditemukan

SQL> alter session set container=cdb$root;

```
SQL> alter session set container=cdb$root;

Session altered.

SQL>
```

Berpindah ke container utama CDB$ROOT

SQL> alter pluggable database pdb5 close immediate;

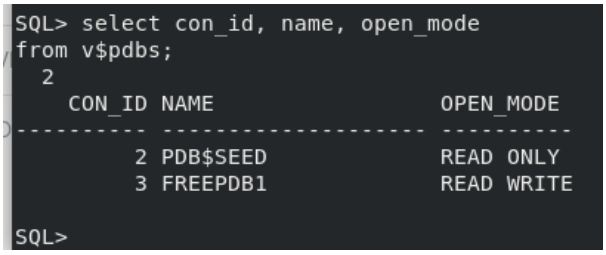
```
SQL> alter pluggable database pdb5 close immediate;
alter pluggable database pdb5 close immediate
*
ERROR at line 1:
ORA-65011: Pluggable database PDB5 does not exist.
Help: https://docs.oracle.com/error-help/db/ora-65011/


SQL>
```

Error terjadi karena PDB5 belum dibuat, sehingga tidak ada pluggable database bernama PDB5 untuk ditutup

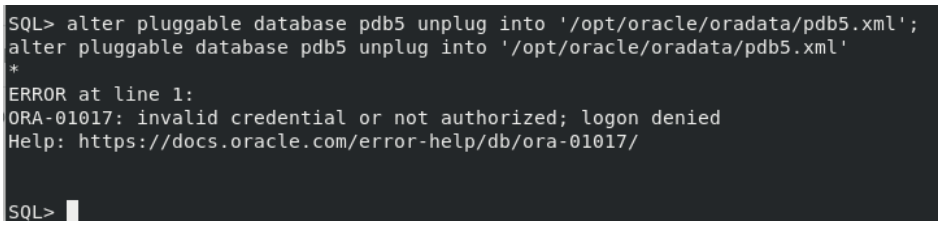
SQL> select con_id, name, open_mode

from v$pdbs;



Menampilkan daftar pluggable database (PDB), ID container dan status opennya dari V$PDBS

SQL> alter pluggable database pdb5 unplug into '/opt/oracle/oradata/pdb5.xml';



Error terjadi karena PDB5 belum dibuat, sehingga tidak dapat dilakukan proses unplug

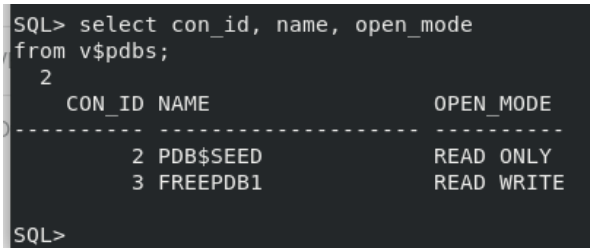
Activities - Files - Other Locations - Computer

/opt/oracle/oradata/FREE/pdb5.xml -- right click - Open with Text Editor

(di skip karena tidak ada file tersebut)

SQL> select con_id, name, open_mode

from v$pdbs;



Menampilkan daftar pluggable database (PDB), ID container dan status opennya dari V$PDBS

SQL> drop pluggable database pdb5 keep datafiles;

```
SQL> drop pluggable database pdb5 keep datafiles;
drop pluggable database pdb5 keep datafiles
*
ERROR at line 1:
ORA-01017: invalid credential or not authorized; logon denied
Help: https://docs.oracle.com/error-help/db/ora-01017/


SQL>
```

Error karena PDB5 belum dibuat, sehingga tidak ada pluggable database bernama PDB5 yang bisa dihapus

SQL> select con_id, name, open_mode

from v$pdbs;

```
SQL> select con_id, name, open_mode
from v$pdbs;
  2
    CON_ID NAME                 OPEN_MODE
---------- -------------------- ----------
         2 PDB$SEED             READ ONLY
         3 FREEPDB1             READ WRITE

SQL>
```

Menampilkan daftar pluggable database (PDB), ID container dan status opennya dari V$PDBS

SQL> create pluggable database pdbtest

using '/opt/oracle/oradata/FREE/pdb5.xml'

FILE_NAME_CONVERT=('/opt/oracle/oradata/FREE/pdb5/',

'/opt/oracle/oradata/FREE/pdbtest/');

```
SQL> create pluggable database pdbtest
using '/opt/oracle/oradata/FREE/pdb5.xml'
FILE_NAME_CONVERT=('/opt/oracle/oradata/FREE/pdb5/',
                   '/opt/oracle/oradata/FREE/pdbtest/');
  2    3    4  create pluggable database pdbtest
*
ERROR at line 1:
ORA-01017: invalid credential or not authorized; logon denied
Help: https://docs.oracle.com/error-help/db/ora-01017/


SQL>
```

Error terjadi karena file XML pdb5.xml belum ada, karena PDB5 belum dibuat atau di unplug, sehingga tidak menemukan sumber untuk membuat PDBTEST

SQL> select con_id, name, open_mode

from v$pdbs;

```
SQL> select con_id, name, open_mode
from v$pdbs;
  2
    CON_ID NAME                 OPEN_MODE
---------- -------------------- ----------
         2 PDB$SEED             READ ONLY
         3 FREEPDB1             READ WRITE

SQL>
```

Menampilkan daftar pluggable database (PDB), ID container dan status opennya dari V$PDBS

SQL> alter session set container=cdb$root;

```
SQL> alter session set container=cdb$root;

Session altered.

SQL>
```

Berpindah ke container utama CDB$ROOT

SQL> alter pluggable database open;

```
SQL> alter pluggable database open;
alter pluggable database open
*
ERROR at line 1:
ORA-65000: missing or invalid pluggable database name
Help: https://docs.oracle.com/error-help/db/ora-65000/


SQL>
```

Error terjadi karena perintah dijalankan tanpa menyebutkan nama PDB yang ingin dibuka

SQL> select con_id, file#, name from V$DATAFILE;

```
SQL> select con_id, file#, name from V$DATAFILE;

    CON_ID      FILE# NAME
---------- ---------- --------------------
         1          1 /opt/oracle/oradata/
                      FREE/system01.dbf

         2          2 /opt/oracle/oradata/
                      FREE/pdbseed/system0
                      1.dbf

         1          3 /opt/oracle/oradata/
                      FREE/sysaux01.dbf

         2          4 /opt/oracle/oradata/

    CON_ID      FILE# NAME
---------- ---------- --------------------
                      FREE/pdbseed/sysaux0
                      1.dbf

         1          7 /opt/oracle/oradata/
                      FREE/users01.dbf
```

```
         3         12 /opt/oracle/oradata/
                      FREE/FREEPDB1/system
                      01.dbf

         3         13 /opt/oracle/oradata/
                      FREE/FREEPDB1/sysaux
                      01.dbf

         3         14 /opt/oracle/oradata/

    CON_ID      FILE# NAME
---------- ---------- --------------------
                      FREE/FREEPDB1/undotb
                      s01.dbf

         3         15 /opt/oracle/oradata/
                      FREE/FREEPDB1/users0
                      1.dbf


11 rows selected.

SQL>
```

Menampilkan daftar datafile, nomor file dan ID container dalam V$DATAFILE

Unplugging and plugging in PDBs (continue)

SQL> show con_name;

```
SQL> show con_name;

CON_NAME
------------------------------
CDB$ROOT
SQL>
```

Menampilkan nama container yang sedang aktif

SQL> alter pluggable database all open;

```
SQL> alter pluggable database all open;

Pluggable database altered.

SQL>
```

Membuka semua pluggable database (PDB) dari CDB$ROOT

SQL> select con_id, name, open_mode

from v$pdbs;

```
SQL> select con_id, name, open_mode
from v$pdbs;
  2
    CON_ID NAME                 OPEN_MODE
---------- -------------------- ----------
         2 PDB$SEED             READ ONLY
         3 FREEPDB1             READ WRITE

SQL>
```

Menampilkan daftar pluggable database (PDB), ID container dan status opennya dari V$PDBS

SQL> alter session set container=pdbtest;

```
SQL> alter session set container=pdbtest;
ERROR:
ORA-65011: Pluggable database PDBTEST does not exist.
Help: https://docs.oracle.com/error-help/db/ora-65011/


SQL>
```

Error terjadi karena PDBTEST belum berhasil dibuat, sehingga PDBTEST tidak ditemukan

SQL> select * from hr.employees;

```
SQL> select * from hr.employees;
select * from hr.employees
                *
ERROR at line 1:
ORA-00942: table or view "HR"."EMPLOYEES" does not exist
Help: https://docs.oracle.com/error-help/db/ora-00942/


SQL>
```

Error karena HR.EMPLOYEES belum dibuat

SQL> alter session set container=cdb$root;

```
SQL> alter session set container=cdb$root;

Session altered.

SQL>
```

Berpindah ke CDB$ROOT

SQL> alter pluggable database pdbtest close immediate;

```
SQL> alter pluggable database pdbtest close immediate;
alter pluggable database pdbtest close immediate
*
ERROR at line 1:
ORA-65011: Pluggable database PDBTEST does not exist.
Help: https://docs.oracle.com/error-help/db/ora-65011/


SQL>
```

Error karena Pluggable Database (PDB) dengan nama PDBTEST tidak ada

SQL> select con_id, name, open_mode

from v$pdbs;

```
SQL> select con_id, name, open_mode
from v$pdbs;
  2
    CON_ID NAME                 OPEN_MODE
---------- -------------------- ----------
         2 PDB$SEED             READ ONLY
         3 FREEPDB1             READ WRITE

SQL>
```

Menampilkan ID Container (CON_ID), nama PDB, dan mode buka (OPEN_MODE) dari Pluggable Database (PDB) yang ada dalam view V$PDBS

SQL> alter pluggable database pdbtest unplug into '/opt/oracle/oradata/pdbtest.xml';

```
SQL> alter pluggable database pdbtest unplug into '/opt/oracle/oradata/pdbtest.x
ml';
alter pluggable database pdbtest unplug into '/opt/oracle/oradata/pdbtest.xml'
*
ERROR at line 1:
ORA-01017: invalid credential or not authorized; logon denied
Help: https://docs.oracle.com/error-help/db/ora-01017/


SQL>
```

Error karena pdbtest tidak ada

SQL> select con_id, name, open_mode

from v$pdbs;

```
SQL> select con_id, name, open_mode
from v$pdbs;
  2
    CON_ID NAME                 OPEN_MODE
---------- -------------------- ----------
         2 PDB$SEED             READ ONLY
         3 FREEPDB1             READ WRITE

SQL>
```

Menampilkan ID Container, Nama PDB, dan Status Buka (Open Mode) PDB

SQL> drop pluggable database pdbtest keep datafiles;

```
SQL> drop pluggable database pdbtest keep datafiles;
drop pluggable database pdbtest keep datafiles
*
ERROR at line 1:
ORA-01017: invalid credential or not authorized; logon denied
Help: https://docs.oracle.com/error-help/db/ora-01017/


SQL>
```

Error karena PDBTEST tidak ada

SQL> select con_id, name, open_mode

from v$pdbs;

```
SQL> select con_id, name, open_mode
from v$pdbs;
  2
    CON_ID NAME                 OPEN_MODE
---------- -------------------- ----------
         2 PDB$SEED             READ ONLY
         3 FREEPDB1             READ WRITE

SQL>
```

Menampilkan ID Container, Nama PDB, dan Status Buka (Open Mode) PDB

SQL> create pluggable database pdbtest1

using '/opt/oracle/oradata/pdbtest.xml'

NOCOPY TEMPFILE REUSE;

```
SQL> create pluggable database pdbtest1
using '/opt/oracle/oradata/pdbtest.xml'
NOCOPY TEMPFILE REUSE;
  2    3  create pluggable database pdbtest1
*
ERROR at line 1:
ORA-01017: invalid credential or not authorized; logon denied
Help: https://docs.oracle.com/error-help/db/ora-01017/


SQL>
```

Error karena file pdbtest.xml tidak ada

Activities - Files - Other Locations - Computer

/opt/oracle/oradata/FREE/

SQL> select con_id, name, open_mode

from v$pdbs;

```
SQL> select con_id, name, open_mode
from v$pdbs;
  2
    CON_ID NAME                 OPEN_MODE
---------- -------------------- ----------
         2 PDB$SEED             READ ONLY
         3 FREEPDB1             READ WRITE

SQL>
```

Menampilkan ID Container, Nama PDB, dan Status Buka (Open Mode) PDB

SQL> alter session set container=pdbtest1;

```
SQL> alter session set container=pdbtest1;
ERROR:
ORA-65011: Pluggable database PDBTEST1 does not exist.
Help: https://docs.oracle.com/error-help/db/ora-65011/


SQL>
```

Error karena PDBTEST1 tidak ada

SQL> alter pluggable database open;

```
SQL> alter pluggable database open;
alter pluggable database open
*
ERROR at line 1:
ORA-65000: missing or invalid pluggable database name
Help: https://docs.oracle.com/error-help/db/ora-65000/


SQL>
```

Error karena nama PDB tidak disebutkan

SQL> select con_id,file#, name from V$DATAFILE;

```
SQL> select con_id,file#, name from V$DATAFILE;

    CON_ID      FILE# NAME
---------- ---------- --------------------
         1          1 /opt/oracle/oradata/
                      FREE/system01.dbf

         2          2 /opt/oracle/oradata/
                      FREE/pdbseed/system0
                      1.dbf

         1          3 /opt/oracle/oradata/
                      FREE/sysaux01.dbf

         2          4 /opt/oracle/oradata/

    CON_ID      FILE# NAME
---------- ---------- --------------------
                      FREE/pdbseed/sysaux0
                      1.dbf

         1          7 /opt/oracle/oradata/
                      FREE/users01.dbf
```

```
         3         12 /opt/oracle/oradata/
                      FREE/FREEPDB1/system
                      01.dbf

         3         13 /opt/oracle/oradata/
                      FREE/FREEPDB1/sysaux
                      01.dbf

         3         14 /opt/oracle/oradata/

    CON_ID      FILE# NAME
---------- ---------- --------------------
                      FREE/FREEPDB1/undotb
                      s01.dbf

         3         15 /opt/oracle/oradata/
                      FREE/FREEPDB1/users0
                      1.dbf


11 rows selected.

SQL>
```

Menampilkan ID Container, Nomor File, dan Nama Datafile